Figure 1.1 A screenshot shows the output from a 3-axis accelerometer in a modern iPhone in the phyphox app (https://phyphox.org/) showing data indicating the owner is walking with their phone in their pocket. Such sensor data enables a digital twin to mirror and interpret real-world motion and behavior in real time.

Gathering measurements is only half the story; the data must be sent somewhere useful. Advances in wireless communications have matched this sensor revolution. Fifth-generation (5G) cellular networks offer high speeds for high-bandwidth data. In contrast, Low-Power Wide Area Networks (LPWANs) like LoRaWAN can transmit over 10 km (compared to Wi-Fi's 50–100 meters) using 10–20 times less power than Wi-Fi. This efficiency enables sensors to run for years on a single battery, making them ideal for small, resource-constrained devices deployed globally.

IoT is the combination of these widely available commodity sensors and actuators ("things") embedded into objects and connected to communications networks. As you move through the book, you'll learn how to wire up a fleet of LoRaWAN-based sensors, route their data through a cloud pipeline, and use it to update a digital twin model in near real time.

1.2.2 Cloud computing

In the past, deploying a digital twin was restricted to large industrial players due to the massive upfront investment required for hardware (data storage, analysis, simulation, and 3D rendering). Today, modern public cloud computing provides the essential infrastructure needed to create and operate sophisticated digital twins without that barrier.

Users can access scalable compute resources on-demand for complex simulations, elastic storage for continuous sensor data streams, and managed services for data processing, analytics, and machine learning. This pay-as-you-go model eliminates the traditional constraints of purchasing expensive servers and specialized software licenses, allowing rapid prototyping and scaling.

Edge computing complements the cloud by processing data closer to where it's generated. For industrial equipment or remote assets, edge devices perform local analytics, filtering, and decision-making in milliseconds, reducing latency for time-critical operations and minimizing bandwidth